

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) Date: 01-03-26
Time: 3 Hrs Max. Marks: 180

01-03-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE ADV_RTA-4_SYLLABUS

MATHEMATICS: **PRESENT WEEK (100%): Straight Lines, Circles, Parabola, Ellipse, Hyperbola**

PHYSICS: **PRESENT WEEK (100%): EMI, Alternating Current, Kinematics, Laws of motion, Work energy**

CHEMISTRY: **PRESENT WEEK (100%): d & f-block, Metallurgy, Carboxylic acids, Amines, Biomolecules**

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THE NARAYANA GROUP

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TIME:3hrs **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 5 – 8)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	4	16
Sec – III (Q.N : 9 – 16)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	8	32
Total				**16**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 17 – 20)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 21 – 24)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	4	16
Sec – III (Q.N : 25 – 32)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	8	32
Total				**16**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 33 – 36)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 37 – 40)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	4	16
Sec – III (Q.N : 41 – 48)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	8	32
Total				**16**	**60**

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MATHEMATICS **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. A variable line L=0 intersects parabola $y = x^2 - x$ at two distinct points A and B such that AB subtends a right angle at the origin. The locus of the mid-points of the chords of the circle $x^2 + y^2 = 1$ which are along the line L=0 is a circle whose center and radius respectively are

A) $(1,1)$ and $\sqrt{2}$ B)(1,1) and 2 C) $\left(\frac{1}{2},\frac{1}{2}\right)$ and $\frac{1}{2}$ D) $\left(\frac{1}{2},\frac{1}{2}\right)$ and $\frac{1}{\sqrt{2}}$

2. In a AABC, vertex A=(4,4), vertex B lies on $L_1 : y-2=0$ and vertex C lies on $L_2 : x-y+4=0$.

If perimeter of ΔABC is minimum then which of following is correct?

A) $B(4,2)$ and $C(0,4)$

B) $B(3,2)$ and $C\left(\frac{4}{3},\frac{16}{3}\right)$

C) Area of ΔABC is 4 square units

D)Minimum perimeter of ΔABC is $6+2\sqrt{5}$

3. Given two circles $C_1 : x^2 + y^2 + 3\sqrt{2}(x+y) = 0$ and $C_2 : x^2 + y^2 + 5\sqrt{2}(x+y) = 0$.Let the radius of a circle which touches C_1, C_2 and common diameter of C_1, C_2 be $\frac{p}{q}$ where p, q are co-prime positive integers then the value of $p^2 + q^2$ equal to

A) 289 B) 226 C) 205 D)19

4. The sum of all integer values of parameter α for which the point $\left(\alpha,(1+\alpha^2)^{-1}\right)$ does not lie outside the triangle formed by the lines $L_1 : 15y = x+1, L_2 : 78y = 118-23x$ and $L_3 : y+2=0$ is equal to

A) 12 B) 25 C) 14 D) 15

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SECTION - II (Maximum Marks : 16)

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For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

5. Let S_1 be a circle passing through (2,0) and (0,2) and having the smallest possible area. S_2 is the circle touching S_1 externally and passing through origin. S_3 is a circle which touches both S_1 and S_2 externally. If the areas of each of S_1, S_2 and S_3 is same then which of the following is/are correct?

A) Radius of a circle which is orthogonal to circles S_1, S_2 and S_3 is $\sqrt{\frac{2}{3}}$

B) Radius of a circle which is orthogonal to circles S_1, S_2 and S_3 is $\frac{2}{\sqrt{3}}$

C) Common tangent to S_1 and S_3 can be $\left(\sqrt{3}-1\right)x-\left(\sqrt{3}+1\right)y-2=0$.

D) Common tangent to S_1 and S_3 can be $y=x$.

6. A circle S having center at point C passes through the point (1, 2) and touches the lines $L_1: 7x-y=5$ and $L_2: x+y+13=0$ at A and B respectively.

Let L_1, L_2 intersect at point P. Then which of the following is/are INCORRECT?

A) Least area of quadrilateral ACBP is 100 sq. units

B) Least radius of S is $\sqrt{50}$

C) Greatest area of quadrilateral ACBP is 200 sq. units

D) Greatest radius of S is 10

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7. In ΔABC, BC=4, AC=AB=$2\sqrt{2}$. A point P is moving inside the ΔABC such that the square of its distance from side BC is half the area of rectangle whose sides are its distances from sides AC and AB. Let D be the center of the locus of P. Then which of the following is/are correct?

A) Locus of P is an ellipse whose length of latus rectum is $\frac{8\sqrt{3}}{9}$

B) Locus of P is a hyperbola with eccentricity $\sqrt{\frac{3}{2}}$

C) Area of quadrilateral ABCD$=\frac{16}{3}$ sq. units

D) Area of quadrilateral ABCD$=\frac{32}{3}$ sq. units

8. Tangent drawn at P (1, 3) on a parabola intersects its tangent at vertex and axis at M (-1, 5) and T respectively. let R (-5, 5) be a point on SP where S is focus of parabola. Then which of the following is/are correct?

A) Slope of directrix of parabola is $\frac{1}{3}$

B) circumcenter of ΔSMP is $\left(-\frac{1}{2},\frac{7}{2}\right)$

C) $(ST)^2-(SM)^2=(PM)^2$

D) T (-3, 7)

SECTION-III (Maximum Marks : 32)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

9. Let Δ_1 be the area of ΔPQR inscribed in an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b>0)$ and Δ_2 be the area of $\Delta P'Q'R'$ whose vertices are the points lying on the auxiliary circle corresponding to the points P, Q, R respectively. If the eccentricity of the ellipse is $\frac{4\sqrt{3}}{7}$ then the value of $\frac{7\Delta_2}{\Delta_1}$ is equal to

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10. Let A_1 and A_2 be the vertices of the conic C_1: $4(x-3)^2+9(y-2)^2-36=0$. Let the locus of a point P which moves in the plane containing A_1 and A_2 such that $|PA_1-PA_2|=3\sqrt{2}$ be another conic C_2. Let D_1 denotes the distance between foci of conic C_2, D_2 denotes the product of perpendicular distances from the points A_1, A_2 upon any tangent drawn to the conic C_2 and D_3 denotes length of tangent drawn from any point on the auxiliary circle of conic C_1 to the auxiliary circle of the conic C_2. Then the value of $\left(\frac{D_1D_2}{D_3^2}\right)^2$ is equal to

11. Two given circles of radii r_1 and r_2 where $\frac{r_1}{r_2}=3+2\sqrt{2}$ touch each other externally. If the locus of the centre of a circle which touches both these circles externally is a conic S with eccentricity p then the value of $\frac{47p^2}{4}$ is equal to

12. An ellipse having its center at (3,4) touches x-axis at O (0,0). If the slope of major axis of ellipse is 1 and λ is length of latus rectum of the ellipse then the value of $\frac{7\lambda^2}{5}$ is equal to

13. The line $3x+y=8$ touches a hyperbola at P (1, 5) and meets asymptotes of hyperbola at A, B. If (1, 1) is the center of hyperbola, $AB=2\sqrt{10}$ and eccentricity= k then the value of $100k^2$ is equal to

14. At a point 'P' on the parabola $y^2=4x$ tangent and normal are drawn. Tangent intersects x-axis at Q and normal intersects parabola at R such that chord PR subtends an angle of 90^0 at vertex of parabola. The value of $\frac{Areaof\,\Delta PQR}{4\sqrt{2}}$ is equal to

15. A line making an angle θ with positive direction of x-axis intersects the parabola $y^2=8x$ at A and B. If AB is a normal chord which subtends a right at the vertex of parabola then the value of $\tan^2\theta$ is equal to

16. A rectangular hyperbola of latus rectum 2 units passes through (0,0) and has (1,0) as its one focus. If the locus of other focus is a circle, then radius of the circle is equal to

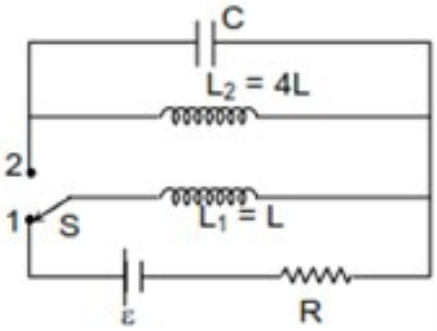
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PHYSICS **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

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Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
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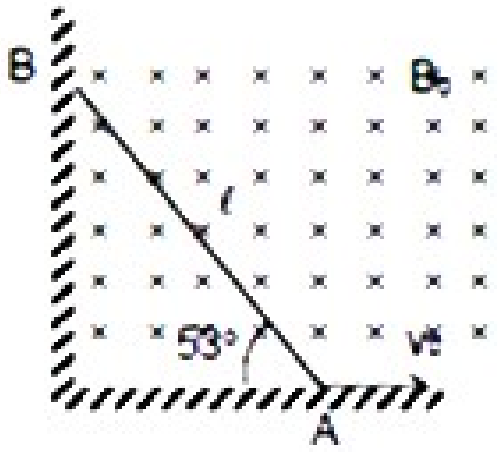
17. In the circuit shown, initially the switch 'S' was getting in position-1 for a long time and then it is shifted to position-2. Initial charge on the capacitor was zero.



The maximum charge on the capacitor is

A) $\frac{\varepsilon}{R}\sqrt{\frac{2CL}{5}}$ B) $\frac{\varepsilon}{R}\sqrt{\frac{4CL}{5}}$ C) $\frac{\varepsilon}{R}\sqrt{\frac{6CL}{5}}$ D) $\frac{\varepsilon}{R}\sqrt{\frac{8CL}{5}}$

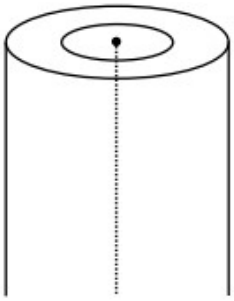
18. The end points of a uniform conducting rod slide on the two mutually perpendicular non-conducting surfaces in a uniform magnetic field as shown. If velocity of end A is V_0 and the rod makes an angle of 53° with the horizontal as shown. Then the maximum value of $|V_A-V_P|$ is : (where P is a point on the rod) (V_A & V_P are potential of point A and P respectively).

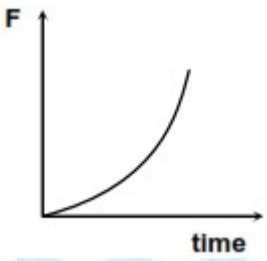


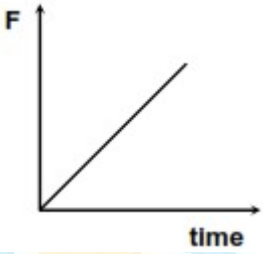
A) $\frac{9}{125}B_0lv_0$ B) $\frac{18}{125}B_0lv_0$ C) $\frac{16}{125}B_0lv_0$ D) $\frac{32}{125}B_0lv_0$

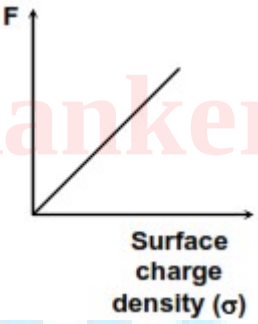
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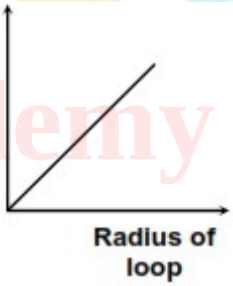
19. An insulating long cylinder is uniformly charged on its lateral surface with surface charge density σ_0. On its flat surface, a circular and conducting loop of radius r is placed coaxially as shown in the figure. There is no friction between the cylinder and the loop. The cylinder is being rotated with a uniform angular acceleration α about its axis. The compressive force in the loop is F and time is t. Which graph is correct?



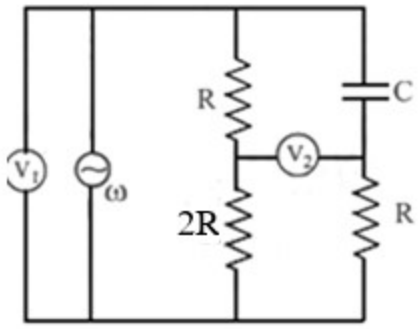
A) 

B) 

C) 

D) 

20. A circuit shown consists of three resistors, one capacitor, 2 ideal AC voltmeters and alternating voltage source of angular frequency ω. Take $R\omega C = 1$. If the reading of V_1 is V, then the reading of V_2 is



A) $\frac{\sqrt{5}V}{2}$ B) $\frac{V}{2}$ C) $\frac{V}{3}\sqrt{\frac{7}{2}}$ D) $\frac{V}{3}\sqrt{\frac{5}{2}}$

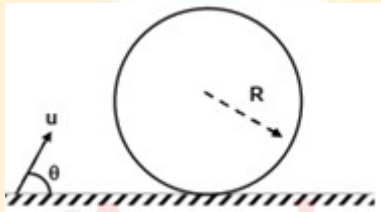
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SECTION - II (Maximum Marks : 16)

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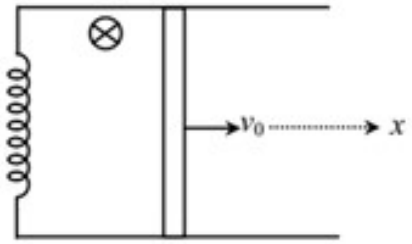
21. A particle is thrown from ground with speed u at an angle θ with horizontal as shown. If u_0 is the minimum value of u so that the particle will cross the horizontal cylinder just grazing at two points on it and θ_0 is the corresponding angle of projection. Then choose the correct option(s).



A) $u_0^2 = Rg\left[2+2\sqrt{2}\right]$ B) $\tan^2\theta_0 = \left[3+2\sqrt{3}\right]$

C) $u_0^2 = Rg\left[2+2\sqrt{3}\right]$ D) $\tan^2\theta_0 = \left[3+2\sqrt{2}\right]$

22. A loop is formed by two fixed parallel conductors connected by a solenoid with inductance L = 2H and a conducting rod of mass m = 8 kg which can freely slide without friction, over the conductors. The conductors are located in a horizontal plane and in a uniform vertical magnetic field B = π Tesla directed into the plane of the paper. The distance between the conductors is $\ell = 2$ m. At the moment t = 0, the rod is imparted an initial velocity v_o = 2 m/s directed to the right. The resistance of loop is negligible. Choose the correct option(s)



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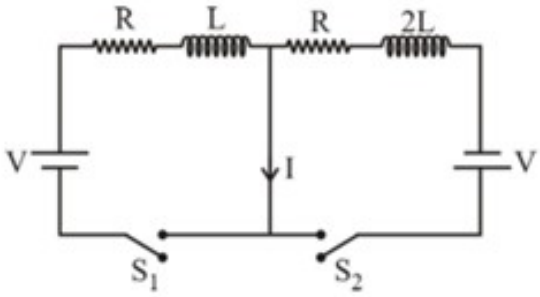
A) The rod will perform periodic motion, but not SHM

B) The rod will perform SHM

C) The time period of oscillation of rod is 4s

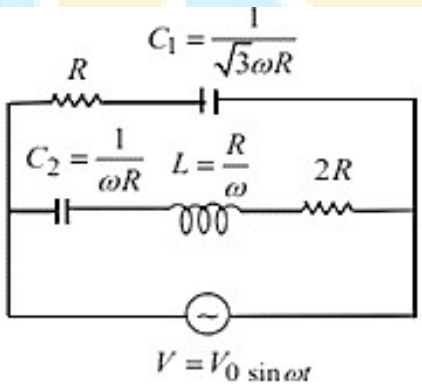
D) The amplitude of oscillation of rod is $\frac{4}{\pi}$ metre.

23. In figure below, the switches S_1 and S_2 are closed simultaneously at t=0 and a current start to flow in the circuit. Both the batteries have the same magnitude of the electromotive force (emf) and the polarities are as indicated in the figure. Ignore mutual inductance between the inductors. The current 1 in the middle wire reaches its maximum magnitude I_{max} at time $t=\tau$. Which of the following statement (s) is (are) true?



A) $I_{max}=\frac{V}{2R}$ B) $I_{max}=\frac{V}{4R}$ C) $\tau=\frac{L}{R}\ell n2$ D) $\tau=\frac{2L}{R}\ell n2$

24. Consider the AC circuit shown and choose correct alternative(s)



A) Effective reactance of the circuit is $\sqrt{3}R$

B) Effective impedance of the circuit is $\frac{2R}{\sqrt{3}}$

C) Current supplied by AC source as a function of time is $\frac{\sqrt{3}V_0}{2R}\sin\left(\omega t+\frac{\pi}{6}\right)$

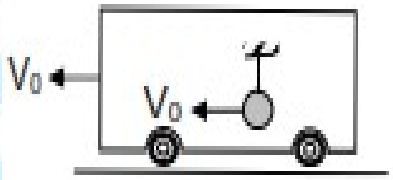
D) Effective wattles current flowing in the circuit is $\frac{\sqrt{3}V_0}{4R}$

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SECTION-III (Maximum Marks : 32)

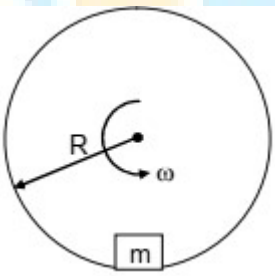
This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**
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25. A pendulum with massless string of length I and bob of mass m is attached inside a railway carriage which is moving with speed v_o as shown. Carriage is brought to rest uniformly by applying brakes. When carriage comes to rest pendulum string is horizontal. Minimum magnitude of retardation of carriage such that bob completes vertical circular motion is (in ms^{-2}). ($g = 10\ ms^{-2}$)



26. A cylinder of radius R is rotating about its horizontal axis with constant $\omega = \sqrt{\frac{5g}{R}}$.

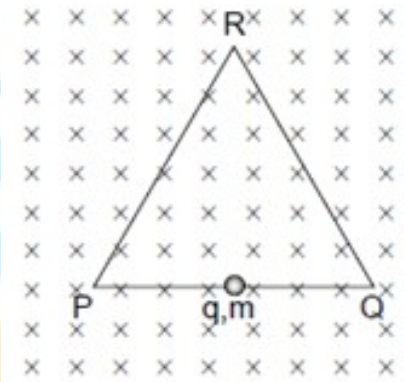
A block of mass is kept on the inner surface of the cylinder. Block is moving in vertical circular motion without slipping. Co-efficient of friction between block and surface of cylinder is μ. If minimum value of μ for complete vertical circular motion of block is $\frac{2\sqrt{6}}{3x}$ then find 'x'.



27. A boat on one bank of the river wants to reach other bank of river flowing with speed v_0. Speed of boat in still water is also v_0. Boat moves in such a way that its distance from a fixed point P remains constant equal to 2d. Time taken to cross the river is $\frac{d}{v_0}\ln\left(x+\sqrt{x^2-1}\right)$, value of x, is (d is the width of the river)

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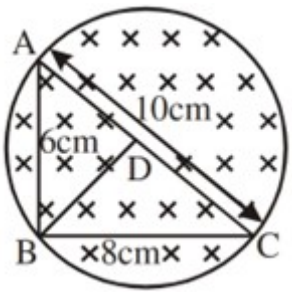
28. A uniform time varying magnetic field $B = B_0t$ is present in a cylindrical region. A fixed wire frame PQR of equilateral triangular shape has slight curve at its corners. Side length of frame is $\ell = 2\sqrt{3}$ m and it is made of insulating material. A charged bead of charge q and mass m can move on the wire frame without friction is released from rest at midpoint of PQ. It is assumed that bead moves through corners of frame without any loss of kinetic energy. The bead moves on the frame and again comes at



initial position in time t_0 and with a speed v_0. It is given that $\frac{qB_0}{m} = \sqrt{3}$ (in Sl unit)

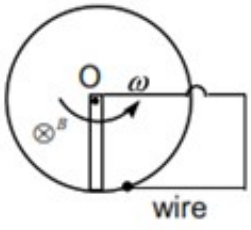
The value of time t_0 (in sec) is……..

29. ABC is a triangle frame made out of a metallic wire. BD is a median of the triangle frame joining the vertex B to the side AC, (also made of the same metallic wire). It is known that AB=6 cm, BC=8 cm and AC=10 cm. The cross-sectional area of the wire is 1 mm^2 and its resistivity is $24\times10^{-9}\Omega m$. The triangle ABC lies in a cylindrical region of magnetic field such that the intersection of the surface of the cylinder with the plane containing the frame forms the circumcircle of triangle ABC. The magnetic field in the region varies at the rate of 0.263 T/s. What is the magnitude of the induced current in mA in the median BD of the frame?

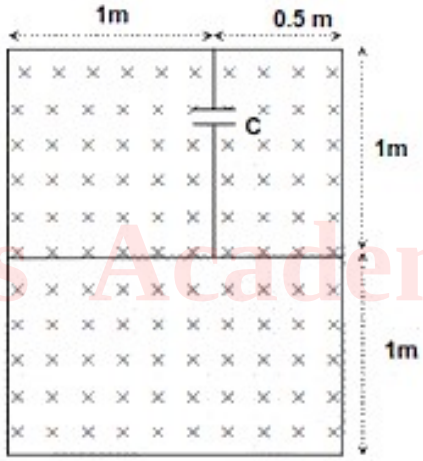


For More Join: @IITJEE_Advanced

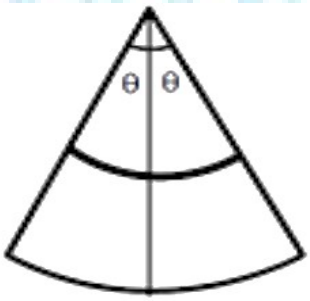
30. A metallic rod of length ℓ and resistance R is free is rotate about one of its ends over a horizontal smooth, rigid circular metallic frame of radius ℓ in an inward magnetic field of induction B. The circuit is completed by means of a conducting wire connected between the center O and the point on the right. If the requires torque to maintain constant angular velocity ω is $\frac{nB^2\ell^2\omega}{R}$, find value of n.



31. Consider a circuit made of uniform wire. A time varying magnetic field $B = -2t\hat{k}\,T$ is switched on in the entire area of the circuit. The charge (In μC) on the capacitor C=7.84 μF in the steady is (The resistance per unit length of the wire is $2\Omega/m$)



32. A rubber band has mass m and Force constant k. When relaxed, the band forms a ring of radius r. The band is placed horizontally on a vertical frictionless cone as shown (angle 2θ is given). At equilibrium position the radius R of the ring formed by the band is $R = r + \frac{mg}{a\pi^2 k}\cot\theta$. Find the value of α



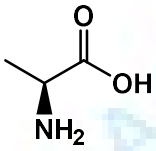
For More Join: @IITJEE_Advanced

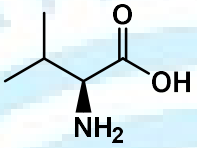
CHEMISTRY **MAX.MARKS: 60**

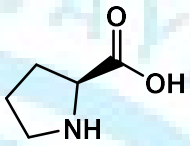
SECTION–I (Maximum Marks: 12)

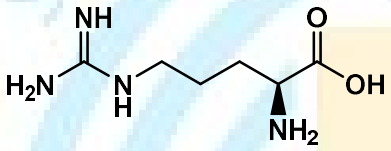
This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

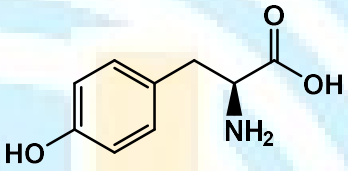
33. Which of the following are non-essential amino acids (considered all are natural amino acids)

(i) 

(ii) 

(iii) 

(iv) 

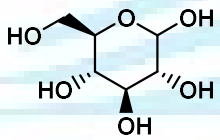
(v) 

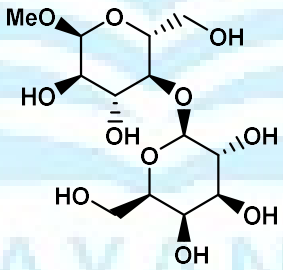
use given information:

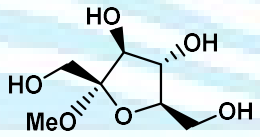
name of amino acid:	Valine	Arginine	Tyrosine	Alanine	Proline
molecular formula:	$C_5H_{11}NO_2$	$C_6H_{14}N_4O_2$	$C_9H_{11}NO_3$	$C_3H_7NO_2$	$C_5H_9NO_2$

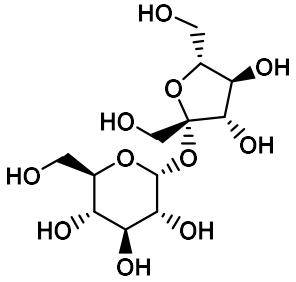
A) (i), (ii) and (v) only
B) (i), (iii), (iv) and (v) only
C) (i) and (v) only
D) (i), (iii) and (v) only

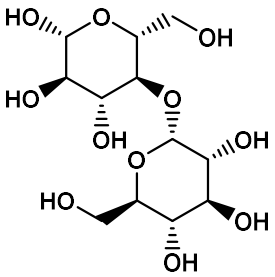
34. The compounds which give positive Fehling's test are

(i) 

(ii) 

(iii) 

(iv) 

(v) 

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Choose the correct answer from the options given below

A) (i) , (ii) and (v) only B)(ii) and (v) only

C) (i) and (v) only D) (iii), (iv) and (v) only

35. Match List-I with List-II

List – I		List – II	
1)	Froth flotation	A)	Oxidises ZnS to ZnO with evolution of SO_2.
2)	Roasting of ZnS concentrate	B)	Separates sulfide minerals from gangue using collectors and frothers.
3)	Calcination of ZnS concentrate	C)	Heating in absence of air to remove volatile impurities; not used for ZnS.
4)	Carbon reduction of ZnO	D)	Reduces ZnO to metallic zinc (zinc vapour) using carbon at high T.
5)	Electrolytic refining of zinc	E)	Produces high-purity zinc by electrolysis of purified $ZnSO_4$ solution.

Choose the correct option

A) 1–B, 2–A, 3–C, 4–D, 5–E B) 1–B, 2–C, 3–A, 4–D, 5–E

C) 1–A, 2–B, 3–C, 4–D, 5–E D) 1–B, 2–A, 3–D, 4–C, 5–E

36. Consider the following sequence of reactions:

i) $4NaCl + K_2Cr_2O_7 + 6H_2SO_4\,(conc.) \rightarrow (A) + 2KHSO_4 + 4NaHSO_4 + 3H_2O$

ii) $(A) + 4NaOH\,(cold, dilute) \rightarrow (B) + 2NaCl + 2H_2O$

iii) $(B) + H_2SO_4 + 2H_2O_2 \xrightarrow{0-5^0\,C,\ either} (C) + Na_2SO_4 + 3H_2O$

The total number of atoms present in one molecule each of (A), (B), and (C) is:

A) 16 B) 17 C) 18 D) 19

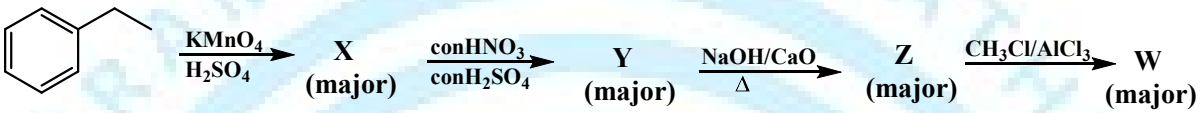
For More Join: @IITJEE_Advanced

SECTION - II (Maximum Marks : 16)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

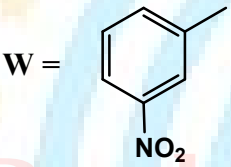
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

37. Choose the correct statement regarding given reaction scheme.



A) Compound 'X' liberates carbon dioxide from $NaHCO_3$

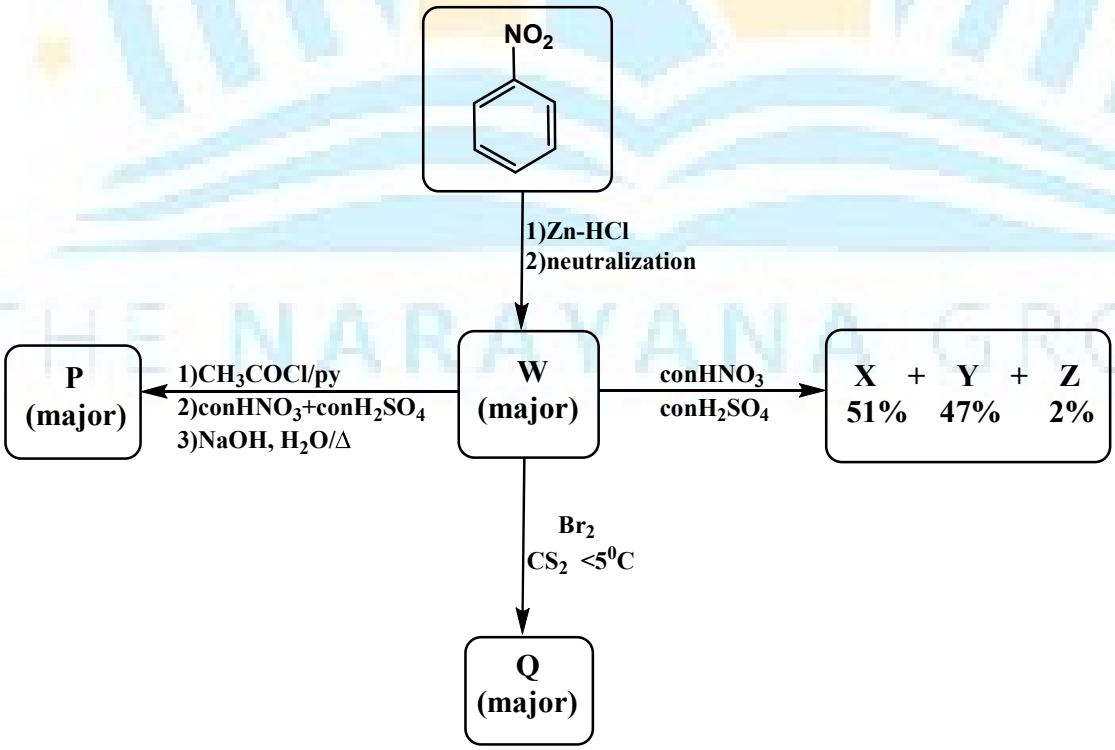
B)



C) Number of sp^2 hybridised carbon atoms in 'Y' are 7

D) The percentage of carbon in compound 'X' is 69 (nearest integer)

38. Choose the correct statement regarding given reaction scheme

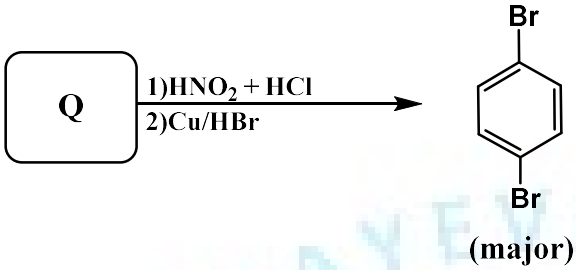


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A) Compound 'P' and Compound 'Y' are positional isomers

B) Compound 'Z' has less basic character than Compound 'P'

C)



(major)

D) Dipole moment of Compound 'Q' is non-zero

39. On a standard Ellingham diagram (ΔG° plotted per mole O_2, $pO_2 = 1$ bar, activities = 1), the formation lines of oxides AO and BO cross at 1200 K. For T < 1200 K the AO line lies below the BO line; for T > 1200 K the BO line lies below the AO line. The CO/CO_2 line intersects the AO line at 900 K and the BO line at 1300 K.

Which of the following statements are correct?

A) At 800 K (standard conditions) metallic A will reduce BO to give AO and free B.

B) At 1400 K (standard conditions) metallic B will reduce AO to give BO and free A.

C) Formation of a volatile oxide ensures the metal will be oxidized at all T because gas formation increases driving force.

D) Since CO/CO_2 lies below AO at 800–900 K, CO will necessarily reduce BO at 1000 K.

40. The correct statements about some of the ions of 3d-series in aqueous solution

A) Cr^{2+} is better reducing agent than Fe^{2+}

B) Co^{3+} is better oxidizing agent than Fe^{3+}

C) When Fe^{3+} is used as oxidizing agent, it attains $3d^7$ electronic configuration

D) Cu is the only metal in 3d series, which is not having ability to displace hydrogen as H_2 from acid (Consider thermodynamic aspect)

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SECTION-III (Maximum Marks : 32)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place;** e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

41. Percentage of carbon present in the product 'P' of the following reaction Scheme is



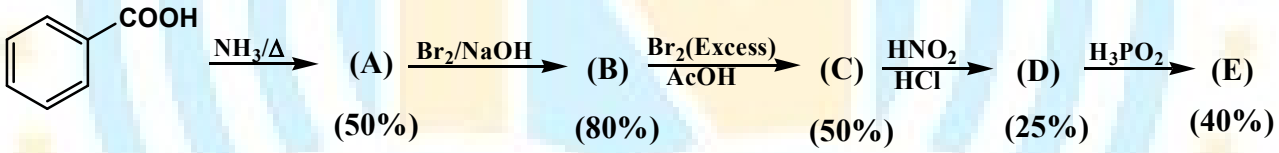
(Use: Molar mass (in g/mol) of H, C, O as 1, 12, 16 respectively)

42. A polyester (Mol. Wt. 418) on complete hydrolysis gives acetic acid (Mol. Wt. 60) and propanoic acid. Acetic acid contributes 35.44% to the total weight of the hydrolysed products. The number of ''CH_3CH_2CO'' groups units Present in the one molecule of polyester is.

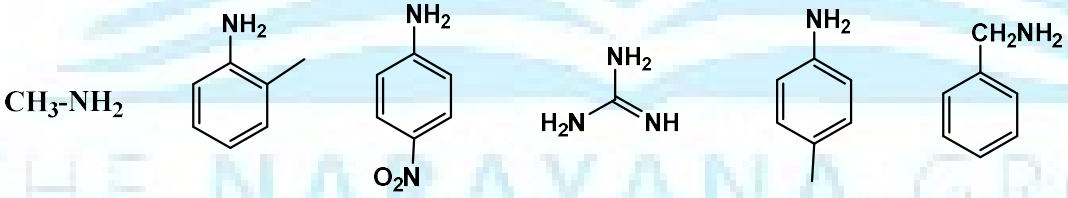
(note: Polyester contains five ester linkages and all are acetyl and propanoyl groups)

43. In the following reaction sequence, the amount of E (in g) formed from 20 moles of benzoic acid is ……

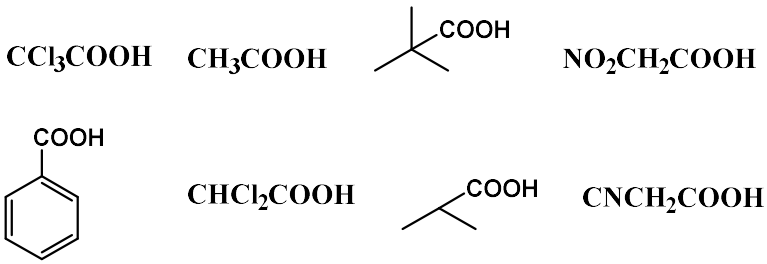
(Atomic weights in g mol^{-}: H=1, C=12, N=14, O=16, Br=80. The yield (%) corresponding product in each step is given in the parenthesis)



44. 'X' is the number of compounds which are more basic than aniline.



'Y' is the number of compounds which are more acidic than formic acid.



Value of (X+Y) is

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45. Consider the following metallurgical and refining processes involving three metals M_1, M_2 and M_3.

Metal M_1:

Dissolves in aerated KCN solution to form a soluble complex $\left[M_1(CN)_2\right]^-$.

It is purified from lead-containing ores using **Cupellation**.

Metal M_2:

Reacts with CO at 330 – 350K to form a volatile tetracarbonyl $M_2(CO)_4$

(Mond process).

Metal M_3:

Reacts with iodine on heating to form a volatile tetraiodide M_3I_4(Van Arkel method).

3d series metal

Let:

A = Total number of gaseous/volatile intermediate species formed during purification of M_2 and M_3 combined.

B = Number of metals from the set $\{M_1, M_2, M_3\}$ that can be refined by Cupellation

C = Number of major impurities among $\{Mn, U, Si, C\}$ oxidized during bessemerization of cast iron.

D = Atomic number of M_3

Calculate: $N = A + B + C + D$

46. A black oxide of manganese reacts with concentrated hydrochloric acid to produce a divalent manganese salt, chlorine gas, and water.

During this reaction:

Manganese is reduced from its highest common oxidation state in that oxide to +2.

Chloride ions are oxidized to chlorine gas.

When **1 mole** of the manganese oxide completely reacts, let:

X = total number of moles of electrons transferred in the reaction

Y = total number of moles of HCl that participate specifically in oxidation (i.e., act as reducing agent)

Find the value of: $N = 5X + Y$.

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47. Permanganate ion (MnO_4^-) undergoes the following transformations:

1) In hot alkaline medium, it forms a green manganate ion.

2) On heating dry $KMnO_4$, oxygen gas is evolved.

3) In acidic medium, it reduces to Mn^{2+}, while in neutral medium (with H_2O_2) it forms MnO_2

4) In neutral medium, 1 mole of $KMnO_4$ produces 1 mole of MnO_2.

Using this information:

Δ = decrease in oxidation number in alkaline conversion

G = moles of gas formed from 2 moles $KMnO_4$ (thermal decomposition)

R = ratio of electrons gained (acidic divided by neutral)

M = mass (in g) of MnO_2 formed from 15.8 g $KMnO_4$

Find: $N = \Delta + G + R + \frac{M}{10}$

(Atomic masses: K=39, Mn=55, O=16)

48. Identify the number of the following processes which are associated with the extraction of zinc from **zinc blende:**

i) Froth floatation method ii) Calcination

iii) Roasting iv) Carbon reduction

v) Electrolytic refining vi) Distillation

vii) Vapour phase refining

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